

Modeling the impacts of structural heterogeneities on the reaction-transport coupling in porous electrodes of lithium-ion battery

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ABSTRACT

Structural optimization of porous electrodes has been a genuinely practical route to improve the energy and power density of lithium-ion battery (LIB). Currently, this relies mainly on the structure-featureless pseudo-two-dimension (P2D) model in theoretical aspect. Herein, a heterogeneous LIB model based on the lattice Boltzmann equations and the corresponding simulation scheme are established to investigate the effects of physical structures on the coupling between ion transport and electrochemical reactions in porous electrodes. The theoretical scheme is validated by comparing the simulated discharging behaviors of lithium manganese oxide/graphite battery systems with that obtained using the P2D model and finite-element method. For battery systems composed of homogeneously distributed structure-uniform active particles, the heterogeneous model gives nearly identical results to that of the P2D model. For battery systems with heterogeneously structured electrodes, however, obviously different behaviors from the prediction of P2D model are obtained, especially at high discharging rates. The P2D model is found to significantly underestimate the structure-sensitive transport-reaction coupling and the non-uniform utilization of active materials, even if the real physical tortuosity based on electrode structure is used. These results highlight the significance of a physical structure-based model and simulation in electrode design for LIB with high capacitance and power density.

HIGHLIGHTS

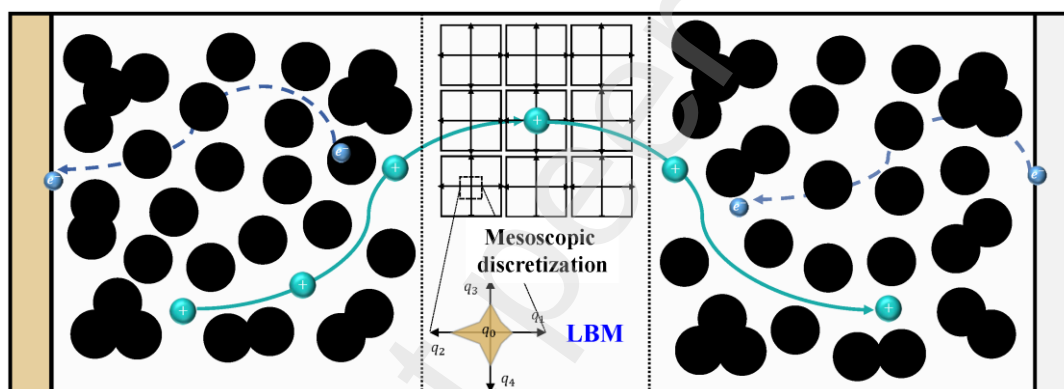
- Heterogeneous 2D model of LIB based on lattice Boltzmann equations.
- Lattice Boltzmann method-based simulation scheme for galvanostatic discharging.
- Effects of microstructures on the coupling between ion transport and electrochemical reactions in porous electrodes.

- Construction of heterogeneously structured porous electrodes using volume exclusion-based random generation method.
- Electrochemical responses and concentration profiles in lithium manganese oxide and graphite battery system.

Keywords

Heterogeneous lithium-ion battery model; Lattice Boltzmann method; Galvanostatic discharging scheme; Constructed electrode structure;

GRAPHICAL ABSTRACT



1. Introduction

The lithium-ion battery (LIB) plays a vital role in the field of electric vehicles. Over the past decade, a large body of research has been dedicated to finding and optimizing LIB electrode materials for increasing energy, power density, and durability[1]. In addition to the structure and composition of activity materials, the structures of the porous electrodes vitally impact the LIB performance[2-5]. Comparing to the materials research, the investigation of the relationship between the electrode structure and battery performance has been relatively inadequate. Recent advance in structure characterization techniques, for example, focused-ion-beam-scanning-electron-microscopy (FIB-SEM)[6, 7], computerized tomography (CT)[8, 9], and so on, provides opportunities to correlate microstructural features and electrochemical responses of porous electrodes. Due to the diverse structural factors in porous electrodes, it is time and labor expansive for a mechanistic understanding of the influences of various microstructural factors on the electrochemical behaviors of porous electrodes merely through experimental characterization[10]. By contrast, theoretical simulations could be an effective and relatively economical approach for resolving the structural effects.

Currently, the mean-field pseudo-two-dimension (P2D) model based on Newman's theory is most widely used to simulate the electrochemistry of porous electrodes[11, 12]. In this theoretical framework, the effects of the structural features on the electrochemical behaviors of porous electrodes are described with an averaged parameter called effective diffusion coefficient, D_{eff} . In the simplest version, the D_{eff} is estimated from electrode tortuosity (τ) through $D_{eff} = \varepsilon/\tau^2$ [13]. Usually, τ is calculated with the overall electrode porosity (ε) by using the empirical Bruggemann relation[14]. Although the ε values are structure-dependent, the same ε values may correspond to

different structures. To better describe the microstructural sensitivity of τ , random walk and homogenization methods have been introduced[15, 16]. The improvement in τ determination can significantly improve the predictability of the P2D model.

In a porous electrode, Li^+ ions undergo coupled diffusion-reaction processes rather than pure diffusion. That is, Li^+ ions in a porous electrode don't necessarily transport along with the full pore depth; instead, they may intercalate into active materials during diffusion. The improved determination of D_{eff} mentioned above well considers the structure sensitivity of the transport processes. Still, it doesn't fully describe the effect of microstructure on the coupling between the transport and electrochemical reactions[17].

To better probe the effects of microstructural heterogeneity on the kinetic coupling between the ions transport and electrochemical reactions in LIB, theoretical simulations should be performed by directly solving the reaction-transport equations in the real rather than averaged electrode structures. There have been a few recent attempts in this regard by using simulation approaches of finite element method (FEM), finite volume method (FVM), and lattice Boltzmann method (LBM)[8, 18, 19]. Since the FEM and FVM are based on continuum transport equations, they are relatively weak in treating the sharp (non-smooth) solid/solution interfaces (boundaries) in the complex porous structure. They, therefore, are not ideal approaches to simulate the effects of the mesoscale heterogeneities on the transport and reaction kinetics[20]. Moreover, the FEM has difficulties maintaining the local conservation for continuous trial functions, while the FVM is difficult to treat the complex irregular meshes accurately. In comparison, the LBM is considerably advantageous in mesoscale simulation because the Boltzmann equation, in principle, is discrete-based, and the distribution functions can easily satisfy local conservation[21].

The application of LBM in the electrochemical system is still at its infant stage. One of the major challenges lies in the implementation of the complex boundary relations between the unknown local variables (potentials and ion concentrations) at solid/electrolyte interfaces and the applied total current density flowing through the battery. To avoid the integration of the exponential Butler-Volmer expression in the iterative solving of LBM equations, Jiang et al.[22], in their recent LBM-based LIB model have used a linear formula between the local current density and interfacial potential difference and assumed a constant charge-transfer resistance at the solid/electrolyte interfaces in the whole electrode domain. This is a significant compromise since the interfacial charge-transfer resistance depends on the local concentrations at the surfaces of solid and electrolyte, which vary significantly in the electrode domain; and the linear formula of current density applies only at small overpotentials, which should not be accurate in real electrode, especially at high charging/discharging rates.

This paper proposes a 2D heterogeneous model of LIB by converting the mass and charge conservation equations into lattice Boltzmann equations (LBEs) through asymptotic expansion. An LBM simulation scheme based on this model is established to investigate the effects of physical structures on the coupling between ion transport and electrochemical reactions in porous electrodes. The discharging behaviors of lithium manganese oxide/graphite battery systems with porous electrode structures were obtained using the volume exclusion-based random generation method. The results show that the P2D model significantly underestimates the structure-sensitive transport-reaction coupling and the non-uniform utilization of active materials, even if the real physical tortuosity based on electrode structure is used.

2. Models and simulations

2.1 Governing equations for LIB porous electrodes

In a porous electrode of LIB, the Li^+ transport in the particles of active materials and the electrolyte contained in the pore domain, and transfer across the interfaces between the active materials and electrolyte phase. The transport of Li^+ in the electrolyte results in heterogeneous profiles of electrostatic potential in the electrolyte phase and the Li^+ concentration in both active materials and the electrolyte phase. Considering that the electric conduction materials are usually added and/or coated on the surface of active materials in the porous electrodes and that the transport of electrons in the electric conduction materials is much faster than that of Li^+ in the electrolyte phase, we may neglect the heterogeneity of electrostatic potential in the electric conduction materials over the entire electrode. As for the active materials, it has been shown that the Li^+ and electron may transport together as a neutral quasi-particle “polaron” in the intercalation materials involving the redox processes of transition metal centers, especially for those with very low electric conductivity. In active materials with high electric conductivity, such as graphite, the electrons should transport much faster than Li^+ , which makes the electrons largely delocalized. In either case, the heterogeneity of electrostatic potential can be neglected in the active potentials. In other words, the solid phase in a porous electrode can be approximately considered iso-potential. Thus, the electrochemical behaviors of a LIB porous electrode can be roughly described by the transport of Li^+ in the solid active materials and electrolyte, and the transfer across the interfaces between the active materials and electrolyte, which obey the mass and charge conservation equations as detailed below.

(1) The mass conservation equation for the transport of Li^+ in solid phase,

$$\frac{\partial c_{\text{Li}^+(s)}}{\partial t} = -\nabla(-D_{\text{Li}^+(s)}\nabla c_{\text{Li}^+(s)}) \quad (1)$$

where $c_{\text{Li}^+(s)}$ and $D_{\text{Li}^+(s)}$ refer to the concentration and diffusion coefficient of Li^+ in solid materials. By using the iso-potential approximation, the transport of Li^+ in the solid phase only involves diffusion.

(2) The mass conservation equation for the transport of Li^+ in electrolyte,

$$\frac{\partial c_{\text{Li}^+(l)}}{\partial t} = -\nabla(-\tilde{D}_{\text{Li}^+(l)}\nabla c_{\text{Li}^+(l)}) + \nabla \cdot \left(\frac{D_{\text{Li}^+(l)}z_{\text{Li}^+}Fc_{\text{Li}^+(l)}}{RT} \nabla \phi_l \right) \quad (2)$$

where $c_{\text{Li}^+(l)}$ and ϕ_l refer to the local concentration of Li^+ and electric potential in the electrolyte, $z_{\text{Li}^+} = 1$ is charge number of Li^+ , and $\tilde{D}_{\text{Li}^+(l)}$ is the so-called chemical diffusivity of Li^+ which is defined to correct the effect of electrostatic coupling between the cations and anions on the mobility of Li^+ in concentrated solution and can be expressed as $\tilde{D}_{\text{Li}^+(l)} = \left(1 + \frac{\partial \ln f_{\text{Li}^+(l)}}{\partial \ln c_{\text{Li}^+(l)}}\right) D_{\text{Li}^+(l)}$, where $D_{\text{Li}^+(l)}$ is the dilute-solution limit diffusion coefficient and $f_{\text{Li}^+(l)}$ is the activity coefficients of Li^+ . In the case that the ideal solutions are used, $f_{\text{Li}^+(l)} \approx 1$ and $\tilde{D}_{\text{Li}^+(l)} \approx D_{\text{Li}^+(l)}$.

The first term on the right side of Eq. (2) represents the concentration change due to the diffusion, while the second term is due to electric field (potential gradient)-driven migration.

(3) The mass conservation equation for the transfer of Li^+ at the solid-electrolyte interface,

$$-\tilde{D}_{\text{Li}^+(l)}\nabla c_{\text{Li}^+(l)} \cdot \hat{n} - \frac{D_{\text{Li}^+(l)}z_{\text{Li}^+}Fc_{\text{Li}^+(l)}}{RT} \nabla \phi_l \cdot \hat{n} = -D_{\text{Li}^+(s)}\nabla c_{\text{Li}^+(s)} \cdot \hat{n} \quad (3)$$

(4) The charge conservation equation for the transfer of e^- at the solid-electrolyte interface,

$$-D_{\text{Li}^+(s)}\nabla c_{\text{Li}^+(s)} \cdot \hat{n} = \frac{i_{\text{local}}}{F} \quad (4)$$

where $\hat{n}$ is the unit vector normal to the interface, i_{local} is the local current density flowing across the solid-electrolyte interfaces, which can be described with the Butler–Volmer equation, that is,

$$i_{\text{local}} = i^0 \left[\exp\left(\frac{\alpha_a F}{RT}(\phi_s - \phi_l - E^{eq})\right) - \exp\left(-\frac{\alpha_c F}{RT}(\phi_s - \phi_l - E^{eq})\right) \right] \quad (5)$$

where E^{eq} refers to the equilibrium (open-circuit) potential established between the solid with state

of charge (SOC) corresponding to the Li^+ at the solid surface at time t ($c_{\text{Li}^+(s)}^{**}$) and the electrolyte with bulk concentration of $c_{\text{Li}^+(l)}^{**}$), $i^0 = nFk^0 c_{\text{Li}^+(l)}^{**\alpha_a/n} (c_{\text{Li}^+(s),\text{max}}^{**} - c_{\text{Li}^+(s)}^{**})^{\alpha_a/n} c_{\text{Li}^+(s)}^{**\alpha_c/n}$ referring to the exchange current density of the interfacial charge transfer reaction at E^{eq} , α_a and α_c are the transfer coefficients of the anodic and cathodic directions of the interfacial charge transfer reaction respectively, n is the reaction charge number, k^0 is the standard reaction rate constant of the interfacial charge transfer reaction, $c_{\text{Li}^+(s),\text{max}}^{**}$ is the saturated lithium concentration in the solid. The term $c_{\text{Li}^+(s),\text{max}}^{**} - c_{\text{Li}^+(s)}^{**}$ in Eq. (5) represents the concentration of the unoccupied Li^+ sites.

Based on the charge conversion, the integral of the local current density over a porous electrode (anode or cathode) is should equal to the current density flowing through the interface between the current collector and the active porous layer, i_{total} , that is,

$$i_{app} = \int_{\Omega} i_{local} d\Omega \quad (6)$$

where Ω represents the electrode space and $d\Omega$ a small volume element.

The mass and charge conservation equations shown so far for Li^+ transport and transfer are not sufficient to solve the profiles of Li^+ concentrations and electrostatic potentials because the three variables ($c_{\text{Li}^+(s)}$, $c_{\text{Li}^+(l)}$ and ϕ_l) are involved in the two main equations (Eq. (1) and Eq. (2)). Eq. (3) and Eq. (4) are the only boundary equations. Therefore, one additional equation relating to these variables is required. This can be obtained from the electroneutrality conditions for electrolyte, that is,

$$c_{\text{Li}^+(l)} z_{\text{Li}^+} + z_a c_{a(l)} = 0 \quad (7)$$

and the mass conservation equation for the transport of anion in electrolyte,

$$\frac{\partial c_{a(l)}}{\partial t} = -\nabla(-\tilde{D}_{a(l)}\nabla c_{a(l)}) + \nabla \cdot \left(\frac{D_{a(l)}z_a F c_{a(l)}}{RT} \nabla \phi_l \right) \quad (8)$$

where $c_{a(l)}$, z_a , and $\tilde{D}_{a(l)}$ refer to the concentration, charge number, and chemical diffusivity of the anion in the electrolyte solution, and $D_{a(l)}$ is the diffusion coefficient of anion in the ideal dilute solution. For a binary z/z electrolyte, $z_a = -z_{Li^+}$ and $c_{Li^+(l)} = c_{a(l)}$. By combining Eq. (2), (7), and (8), one has the charge conservation equation associated with the electroneutrality condition in the electrolyte, that is,

$$\nabla \cdot (\sigma \nabla \phi_l) + \nabla(z_{Li^+}(\tilde{D}_{Li^+(l)} - \tilde{D}_{a(l)})F\nabla c_{Li^+(l)}) = 0 \quad (9)$$

where $\sigma = \frac{z_{Li^+}^2 F^2 c_{Li^+(l)}(D_{Li^+(l)} + D_{a(l)})}{RT}$ is the conductivity of the electrolytic solution. It can be proved that $\tilde{D}_{Li^+(l)} - \tilde{D}_{a(l)} = -\frac{2RT\sigma}{c_{Li^+(l)}F^2} \left(1 + \frac{\partial \ln f_{\pm}}{\partial \ln c_{Li^+(l)}} \right) (1 - t_+)$, $f_{\pm}$ is the geometric mean activity coefficients of cation and anion, t_+ is the transference number of cation, and the details please refer to the literature[23]. The second term in the left side of Eq. (9) represents the charge accumulation due to the diffusivity difference between the cations and anions.

Because that no anion enters the solid phase, one has the charge conservation equation for the transport of anion at the solid-electrolyte interface,

$$(-z_a F \tilde{D}_{a(l)} \nabla c_{a(l)}) + \left(\frac{D_{a(l)} z_a^2 F^2 c_{a(l)}}{RT} \nabla \phi_l \right) = 0 \quad (10)$$

By combining Eq. (3), (7), and (10), one has the charge conservation equation associated with the electroneutrality condition,

$$(-\sigma \nabla \phi_l - z_{Li^+} F (\tilde{D}_{Li^+(l)} - \tilde{D}_{a(l)}) \nabla c_{Li^+(l)}) \cdot \hat{n} = i_{local} \quad (11)$$

Thus, the distributions of Li^+ concentrations in active particles and electrolytes, and the electrostatic potentials in the electrolyte can be obtained by solving Eq. (1), Eq. (2), and (9) with the boundary conditions of Eq. (3), (4), (5), (6) and (11). In the P2D models, which are rooted in Newman's theory for porous electrodes, the above equations are not directly solved in the real space

of porous electrodes. Instead, the volume-averaged tortuosity (τ) is used to represent the structural features so that the electrochemical responses can be described by the transport of Li^+ in continuous straight pores and that in spherical solid particles homogeneously distributed along the ion transport path, which couple to each other through charge transfer reactions at the solid-electrolyte interface. As stated in Introduction, this mean-field method cannot capture great heterogeneity of the concentration and potential distributions under high discharging/charging rates.

2.2 Lattice Boltzmann Equations for porous electrodes of LIB

2.2.1 Governing equations

The key in LBM simulations is to transform the governing equations of the studying system into the form of lattice Boltzmann equations (LBE), which is the discrete form of the Boltzmann equation (BE) originally developed to describe the evolution of particle distribution function (PDF) as a function of particle velocities ($\mathbf{v}$) and coordinates ($\mathbf{x}$), namely, $f(\mathbf{x}, \mathbf{v}, t)$, with time (t) under the collision and the force-driven drift. As well as discretizing the coordinate space of the studying system into microscopic lattices in which the thermodynamic laws are assumed to apply, the core idea in LBM simulation is to discretize the continuous velocity space of the system ($\mathbf{v}$) into a small number of discrete velocities (ξ_i) toward different directions. Accordingly, the PDF in each lattice is discretized into fractions assigned to different velocity directions, that is, $f_i(\mathbf{x}, t) = \frac{w_i}{\theta(\xi_i)} f(\mathbf{x}, \xi_i, t)$, where w_i and ξ_i are the weight factor and velocity in the direction i , $\theta(\xi_i)$ is a factor added to satisfy the Gauss-Hermite rule for converting the integral term into the sum of discretized terms, which guarantees the mass and momentum conservation in the velocity discretization process, and in the meantime the validity of the macroscopic equations.

The lattice velocity space is mostly discretized with the DnQm scheme, in which Dn means

that the spatial dimension of the LBE model is n , and “Qm” means that the discrete number of velocity directions is m . Furthermore, for 2D problems, the D2Q5 model is commonly used to balance the computation cost and accuracy (**Fig. 1a**)[24]. Based on the Gauss-Hermite quadrature rule, the discretized velocities and weight factors for different velocity directions in the D2Q5 model are as follows,

$$\xi_i = \begin{cases} (0,0) \\ \xi_s \left(\cos \left[(i-1) \frac{\pi}{2} \right], \sin \left[(i-1) \frac{\pi}{2} \right] \right) \end{cases} \quad w_i = \begin{cases} \frac{1}{3}, & i = 0 \\ \frac{1}{6}, & i = 1 \sim 4 \end{cases} \quad (12)$$

where ξ_s is the streaming speed, it is calculated by the equation: $\xi_s = \Delta x / \Delta t$, Δx and Δt are the mesh size and time step, respectively.

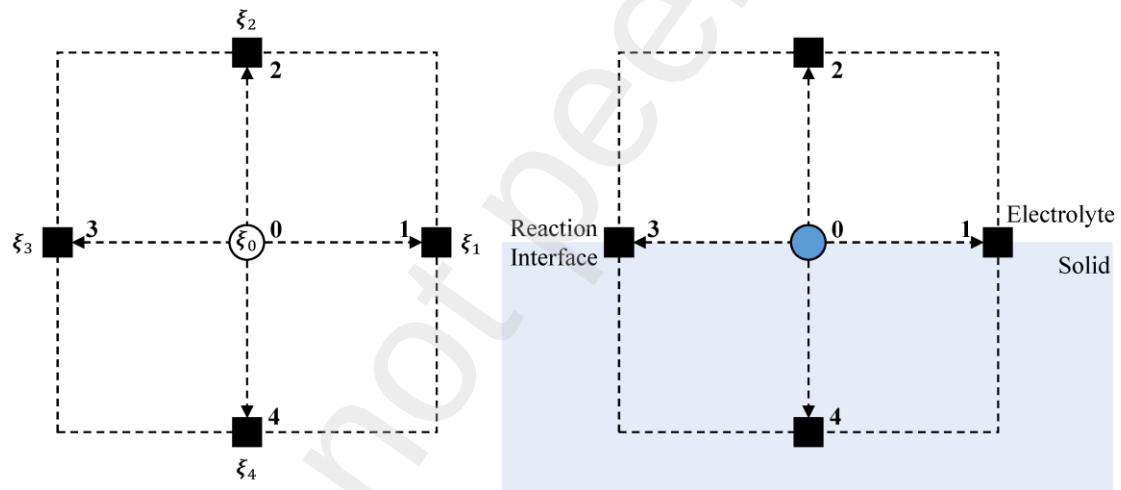


Fig. 1. Schematic illustration of the space and velocity discretization of D2Q5 lattices: (a) in single phases; (b) at the interface.

A major task in the LBM is to simplify the collision term, which in the original BE appears as a complicated and cumbersome double integral over velocity space to include all the possible outcomes of two-particle collisions. In the most successful Bhatnagar-Gross-Krook (BGK) approach, the collision-induced changes of $f_i(\mathbf{x}, t)$ are treated with a linear relaxation toward equilibrium, that is, $(\partial f_i(\mathbf{x}, t) / \partial t)_c = \frac{1}{\tau} (f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t))$, where τ is the relaxation factor,

and $f_i^{eq}(\mathbf{x}, t) = w_i \sum f_i(\mathbf{x}, t)$ referring to the equilibrium partial distribution function when convection is not considered. Despite its simplicity, the BGK method has been proved to effectively maintain the mass and momentum conservation in the collision processes. It is essential for microscopic simulation of particle dynamics and has been a challenge in developing the LBM.

The discretization of velocity space thus converts the BE into a number of LBEs describing the evolution of PDF in different directions, which can be expressed by the following generalized form[25],

$$f_i(\mathbf{x} + \xi_i \Delta t, t + \Delta t) - f_i(\mathbf{x}, t) = -\frac{1}{\tau} \left(f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t) \right) + w_i F \Delta t + w_i S \Delta t \quad (13)$$

More specifically, it says that the partial PDF in lattice $\mathbf{x}$ at time t , $f_i(\mathbf{x}, t)$, becomes the partial PDF in the lattice $\mathbf{x} + \xi_i \Delta t$ at time $t + \Delta t$, $f_i(\mathbf{x} + \xi_i \Delta t, t + \Delta t)$. The first, second, and third terms on the right side of Eq. (13) refer to the change of PDF due to particle collision, field-induced drift, and source which produces particles, respectively.

Although the LBM was originally developed to deal with the transport processes of particles (molecules), it could solve the dynamic distribution of any macroscopic physical quantities described with partial differential equations that can be written the form of particle transport[26]. Specific to the problems in this work, Eq. (1) and (2) are the transport equations for Li^+ in solid and electrolyte respectively, therefore can be directly solved with LBM by substituting $c_{\text{Li}^+(s)}$ and $c_{\text{Li}^+(l)}$ respectively for the distribution function, f . As for Eq. (9), it can be considered a steady-state transport equation of electric-field upon substituting ϕ_l^* for f , with $\partial \phi_l / \partial t = 0$. Considering that the electric-field transports much more rapidly than the ions, we can decouple the electric-field and concentration field when implementing the numerical solutions of these equations, more specifically, assuming one is invariable in the time intervals when computing the other. Thus,

the charge associated with the diffusivity difference between cations and anions in Eq. (9) can be treated as a source term of an electric field; and the electromigration term in Eq. (2) can be treated as the drift caused by the force due to electric field[27]. **Table 1** lists the distribution functions, forces, and sources in the LBEs corresponding to Eq. (1), (2), and (9), respectively (derivation details are given in section S1 in the Supplementary information).

Table 1 Values of f , F , S

Governing equations	f	F	S
Eq. (1)	$c_{\text{Li}^+(s)}$	0	0
Eq. (2)	$c_{\text{Li}^+(l)}$	$-\left(1 - \frac{1}{2\tau}\right)\xi_i \frac{z_{\text{Li}^+} F c_{\text{Li}^+(l)}}{RT} \nabla \phi_l$	0
Eq. (9)	ϕ_l	0	$-\nabla \cdot (z_{\text{Li}^+} (\tilde{D}_{\text{Li}^+(l)} - \tilde{D}_{a(l)}) \nabla c_{\text{Li}^+(l)})$

The macroscopic equations of Eq. (1), (2), and (9) can be fully recovered from the corresponding LBEs through asymptotic expansion, which gives the expressions of the corresponding relaxation times (section S1 in the Supplementary information), that is,

$$\tau_\chi = 3D_\chi \Delta t_\chi / \Delta x^2 + 0.5 \quad (14)$$

where χ represents the transport processes of Li^+ in solid, Li^+ in electrolyte, or electric field in electrolyte, and $D_\chi = D_{\text{Li}^+(s)}$, $\tilde{D}_{\text{Li}^+(l)}$ or $\sigma\theta$ accordingly, θ is the coefficient calculated by the Nernst-Einstein equation, have a value of $1.08 \times 10^{-7} \text{ m}^3 \text{ s}^{-1} \text{ S}^{-1}$, Δt_χ refers to the time steps for the corresponding processes. In our simulations, the physical space step, Δx , is set as the grid resolution used to discretize the electrode, namely, 1 μm . The values of D_χ and Δt_χ , and the corresponding relaxation factor τ_χ , are listed in **Table 2**.

Table 2 Physical parameters and corresponding dimensionless lattice values for the simulation

Parameter	Physical Value	Lattice Value
$D_{\text{Li}^+(s),a}$	$3.9 \times 10^{-14} \text{ m}^2 \text{ s}^{-1}$	3.9×10^{-5}
$D_{\text{Li}^+(s),c}$	$1 \times 10^{-13} \text{ m}^2 \text{ s}^{-1}$	1×10^{-4}
$\tilde{D}_{\text{Li}^+(l)}$	$7.5 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$	7.5×10^{-2}
σ	$1.54 \times 10^{-2} \text{ S cm}^{-1}$	1.66×10^{-1}
$\Delta t_{\text{Li}^+(s)}$	$1 \times 10^{-3} \text{ s}$	1
$\Delta t_{\text{Li}^+(l)}$	$1 \times 10^{-3} \text{ s}$	1
Δt_{σ}	$1 \times 10^{-6} \text{ s}$	1
$\tau_{\text{Li}^+(s),a}$	0.5001	0.5001
$\tau_{\text{Li}^+(s),c}$	0.5003	0.5003
$\tau_{\text{Li}^+(l)}$	0.7250	0.7250
τ_{σ}	1	1

Note: It is usually necessary to have $\tau > 0.5$ in LBM simulations otherwise the f_i would oscillate around f_i^{eq} with an exponentially increasing amplitude. Therefore, considering the computational cost and convergence, the dimensionless time-step of 1 in LBM should correspond to a physical time-step of $1 \times 10^{-3}\text{s}$.

2.2.2 Boundary equations

To solve LBEs associated with Eq. (1), (2), and (9), the corresponding boundary equations should be discretized too. In this work, the bounce-back method proposed by Kang et al. was utilized[28]. **Fig.1b** illustrates the boundary at the reactive interface. Correspondingly, the PDF boundary conditions of Eq. (3), (4), and (11) have the following discrete forms[24],

$$\xi_s^{\text{Li}^+} \left(f_4^{c_{\text{Li}^+(s)}}(\mathbf{x}_b, t + \Delta t) - f_2^{c_{\text{Li}^+(s)}}(\mathbf{x}_b, t) \right) = \frac{i_{\text{local}}}{F} \quad (15)$$

$$\xi_l^{\text{Li}^+} \left(f_4^{c_{\text{Li}^+(l)}}(\mathbf{x}_b, t) - f_2^{c_{\text{Li}^+(l)}}(\mathbf{x}_b, t + \Delta t) \right) = \frac{i_{\text{local}}}{F} \quad (16)$$

$$\xi_l^{\phi_l} \left(f_4^{\phi_l}(\mathbf{x}_b, t) - f_2^{\phi_l}(\mathbf{x}_b, t + \Delta t) \right) = i_{\text{local}} \quad (17)$$

where $\mathbf{x}_b$ is the boundary coordinates, and $\xi_s^{\text{Li}^+} = \Delta x / \Delta t_{D_{\text{Li}^+(s)}}$, $\xi_l^{\text{Li}^+} = \Delta x / \Delta t_{D_{\text{Li}^+(l)}}$, $\xi_l^{\phi_l} = \Delta x / \Delta t_{\sigma}$, are the streaming speeds associated with the corresponding transport processes.

In addition, Eq (5) and (6) are implemented with the following discrete form,

$$i_{\text{app}} = \sum_m i_m = \sum_m i_m^0 \cdot \left[\exp \left(\frac{\alpha_a F}{RT} (\phi_s^* - \phi_{l,m} - E_m^{\text{eq}}) \right) - \exp \left(\frac{-\alpha_c F}{RT} (\phi_s^* - \phi_{l,m} - E_m^{\text{eq}}) \right) \right] \quad (18)$$

where m represents the discretized areas of the reactive interface, $i_m^0 = n F k^0 c_{\text{Li}^+(l)}^{\frac{\alpha_a}{n}} \left(c_{\text{Li}^+(s),\text{max}}^{**} - c_{\text{Li}^+(s)}^{*(m)} \right)^{\frac{\alpha_a}{n}} c_{\text{Li}^+(s)}^{*(m)\frac{\alpha_c}{n}}$ referring to the exchange current density corresponding to the Li^+ concentration in the solid at m ($c_{\text{Li}^+(s)}^{*(m)}$). As stated in section 2.1, the solid phase in a porous electrode can be approximately considered iso-potential, that is, ϕ_s^* is invariant with m .

2.3 Model battery systems and Computations

2.3.1. Model battery system

In this paper, we consider a model battery system with lithium manganese oxide ($\text{Li}_y\text{Mn}_2\text{O}_4$) and graphite as the active materials of the cathode and anode, respectively. The corresponding electrochemical reactions are as follows and the material properties parameters can be found in section S2 in the Supplementary information.



The equilibrium potential of the electrode reactions[29], that is,

$$\begin{aligned}
E_{\text{anode}}^{eq}(y) = & 0.7222 + 0.1387 \times y + 0.029 \times y^{0.5} - 0.0172 \times y^{-1} \\
& + 0.0019 \times y^{-1.5} + 0.2808 \times e^{(0.9-15 \times y)} \\
& - 0.7984 \times e^{(0.4465 \times y - 0.4108)}
\end{aligned} \tag{21}$$

$$\begin{aligned}
E_{\text{cathode}}^{eq}(y) = & 4.1983 + 0.0566 \times \tan(-14.5546 \times y + 8.6094) \\
& - 0.02754 \times ((0.9984 - y)^{-0.4925} - 1.9011) \\
& - 0.1571 \times e^{(-0.04738 \times y^8)} \\
& + 0.8102 \times e^{(-40 \times (y - 0.1339))}
\end{aligned} \tag{22}$$

where $y = c_{\text{Li}^+(s)}^{**} / c_{\text{Li}^+(s),\text{max}}^{**}$.

2.3.2. Nondimensionalize of variables and parameters

For convenience, the LBEs are solved with dimensionless lattice quantity. The spatial coordination x space and time t are nondimensionalized by the corresponding physical space-step (Δx) and time-steps (Δt_x), respectively. Thus, the space and time steps in LBM have a value of 1. The corresponding dimensionless diffusivity in LBM is determined according to the following relation[30],

$$\left(\frac{D_x \Delta t_x}{\Delta x^2} \right)_{\text{LBM}} = \left(\frac{D_x \Delta t_x}{\Delta x^2} \right)_{\text{phys}} \tag{23}$$

The corresponding dimensionless values for various parameters are also given in **Table 2**. In addition, the concentrations of Li^+ are nondimensionalized with the characteristic concentration of 14870 mol m^{-3} , the initial concentration of Li^+ in the anode, the voltage are normalized by 1 V, and the temperatures are normalized by 1 K.

2.3.3. The iterative processes

To solve the above LBEs. Firstly, a primary current distribution is utilized as initial conditions, namely, the concentration and electric field are uniformly distributed in the electrodes with the value

of $c_{\text{Li}^+(s)}^0$, $c_{\text{Li}^+(l)}^0$ and ϕ_l^0 . By solving the Eq. (18), the unknown variables ϕ_s^* , i_{local} can be obtained; then by combining Eq. (1), Eq. (2) and Eq. (9) with Eq. (15), (16), and (17), we can obtain the unknown variables $c_{\text{Li}^+(s)}^*$, $c_{\text{Li}^+(l)}^*$ and ϕ_l^* at each iteration step in turn. It's worth noting that the potential distribution is iterated repeatedly till a steady-state is achieved under each concentration distribution. In the paper, the convergence criterion of ϕ_l^* is that the error per time step $Re_{\phi_l^*}$ is much less than the magnitude of the overpotential. Finally, the iteration continued until the voltage for the battery discharge was reduced to 3.4V.

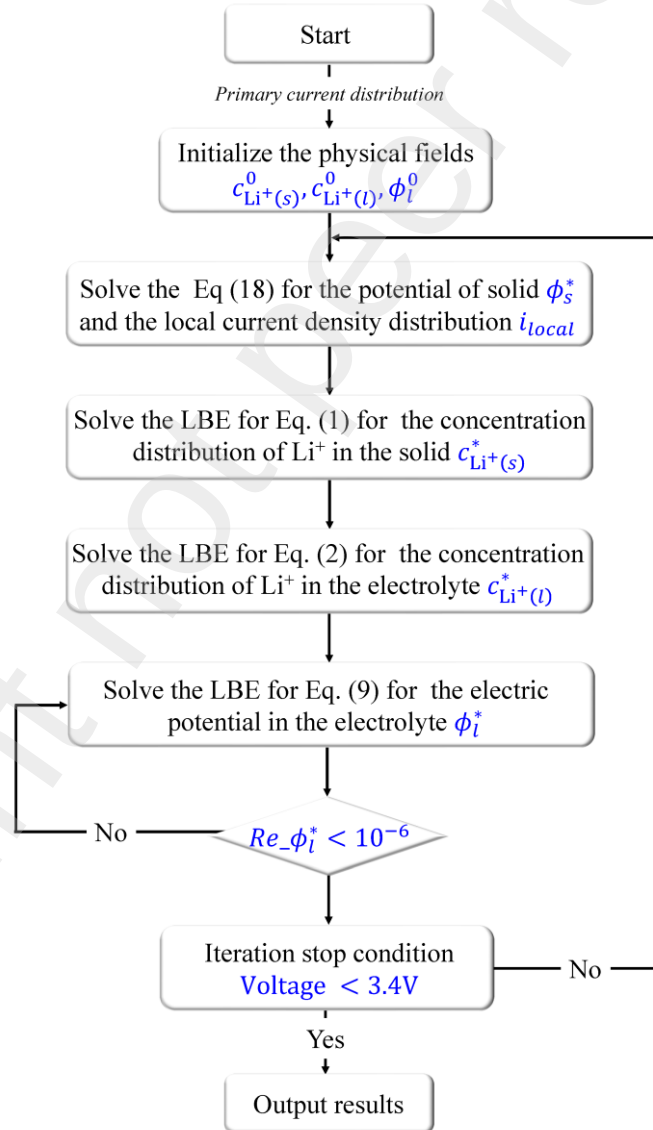


Fig. 2. Schematic of iteration process of the mesoscale theoretical model for LIB.

2.4 Electrode structures

As shown in **Fig. 3**, three electrode structures were used in this paper, and they all are 100×250 lattice units (matrixes) in size and 0.5 in porosity. The structures in **Fig. 3a** and **Fig. 3b** are constructed to validate the model in calculation method and structure treatment, respectively. In **Fig. 3a**, the structure is constructed with continuous rod-like sheet particles so that the electrodes contain straight pores which allows the direct numerical simulations using FEM without P2D approximation. In **Fig. 3b**, the structure is constructed with uniform spherical particles ($7 \mu\text{m}$) that are uniformly distributed in the electrolyte, which should satisfy the hypothetical structure of homogenization of the P2D model. In addition, as shown in **Fig. 3c**, to investigate the effects of microstructural heterogeneities, a heterogeneous structure is constructed using a volume exclusion-based random algorithm, with two main control parameters, namely, porosity and particle size distribution, and the overlap of particles is allowed to control the porosity of electrodes. The generation procedure follows the steps below: (i) Calculating the numbers of particles according to the particle shape and size distributions, and the electrode size and porosity. (ii) Generating the matrixes representing the particles, and randomly placing in the structure space, then searching for other particles around the particle and calculating the overlap rate. The particle will become part of the structure if the overlap rate exceeds the threshold value. (iii) Repeat generating process of (ii) until the desired porosity is met. It should be noted that there is actually a virtual conductive network of electrons in all of the above electrodes but it is not shown in **Fig. 3**.

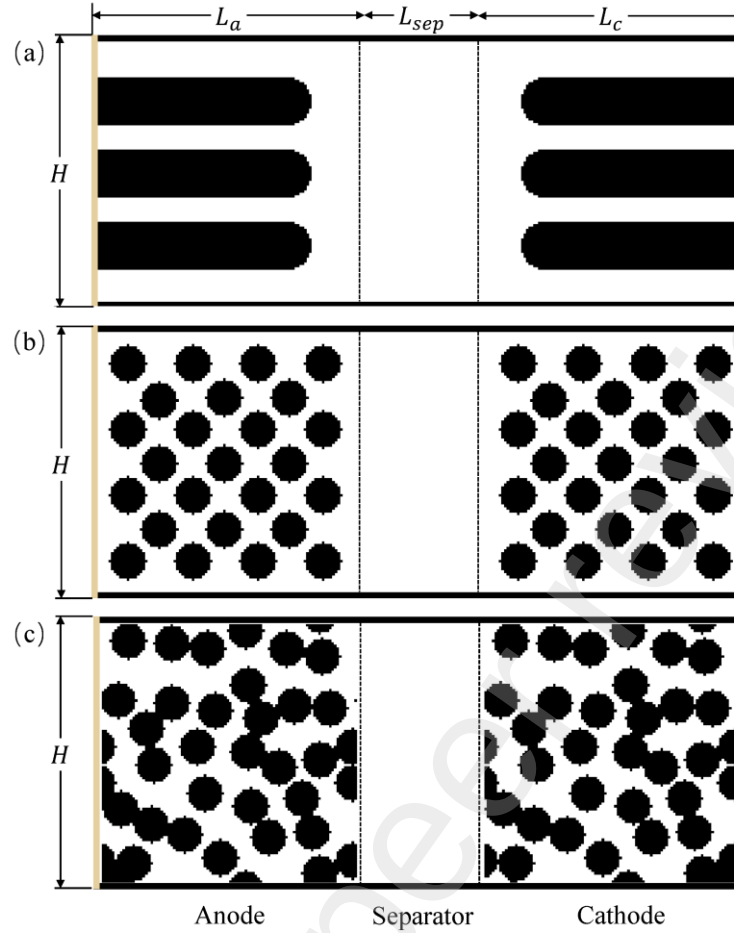


Fig. 3. Electrode composed of (a) structure-uniform rod-like sheet particles, (b) homogeneously distributed structure-uniform spherical particles, and (c) heterogeneously distributed spherical particles.

3. Results and discussion

3.1 Validation of the model and simulation scheme

To validate the heterogeneous model proposed in the paper, we perform LBM simulations on battery systems with electrode structures shown in **Fig. 3a** and **Fig. 3b** by solving the aforementioned LBEs using the software of MATLAB R2021a, and the results are compared with that obtained by using the widely accepted FEM software packages of COMSOL Multiphysics. First, we have compared the simulation results of the heterogeneous model with the FEM simulation in COMSOL Multiphysics for a simply structured array electrode (**Fig. 3a**), for which the FEM can be directly applied without any structure approximation and simplification, at different C-rates. The

simulated voltage responses at different discharge rates are shown in **Fig. 4**. As seen in **Fig. 4a**, the voltage responses with respect to the depth of discharge (DOD) at various discharging rates given by the heterogeneous model are nearly identical to that obtained by the FEM simulation, which indicates the heterogeneous model and the lattice Boltzmann numerical method have the similar correctness to the FEM model in COMSOL Multiphysics. In addition, we have also compared the simulation results of our heterogeneous model with the P2D simulation in COMSOL Multiphysics for a homogeneously structured sphere electrode (**Fig. 3b**), for which the FEM is difficult to simulate directly, and volume averaging structure approximation is needed. The electrode in **Fig. 3b** is an ideal electrode that can be accurately described by the P2D model. It is expected that the heterogeneous model should give the same simulation results as that predicted by the P2D model for the homogeneous electrode structure in **Fig. 3b**. As shown in **Fig. 4b**, the voltage responses obtained by the heterogeneous model agrees well with that given by the P2D model at low C-rates, indicating the correctness of the heterogeneous model and the lattice Boltzmann numerical method again. However, a slight divergence occurs at relatively high C-rates at the late stage of the discharge. This should be mainly due to the small difference between homogeneous electrode structure and P2D assumption. In the P2D model, the electrode structure is perfectly homogeneous, the particle surface Li^+ concentration is uniform. While it is impossible in the homogeneous electrode structure, there is a slight Li^+ concentration gradient on the particle surface due to the transport in the direction of electrode thickness. As the discharge rate increases, there is the increasing importance of lithium ions transport in determining the performance, thus the difference in surface concentration is not negligible at high C-rates.

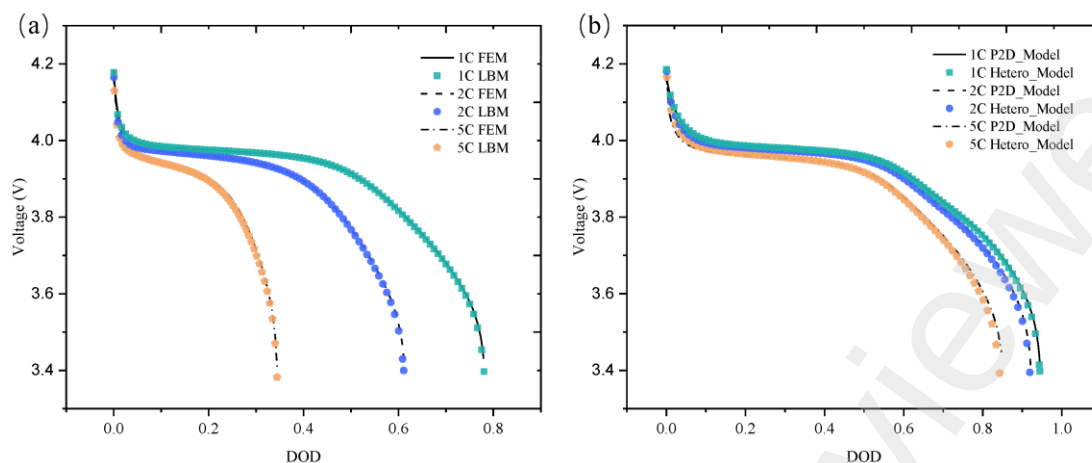


Fig. 4. Comparison of voltage response curves at 1C, 2C, and 5C. The results in (a) are obtained by the FEM and LBM with the same structure; The results in (b) are obtained by the FEM (P2D_Model) and LBM (Hetero_Model) with different structure. The P2D_Model represents the virtual ideal homogenized structure, and Hetero_Model represents the constructed homogeneous structure corresponding to the virtual structure.

3.2 Simulations on the heterogeneously structured electrodes

3.2.1 Discharge rates-dependent utilization of active materials in heterogeneously structured electrodes

Fig.5 shows the simulated concentration profiles of Li^+ in the electrodes in **Fig. 3c** using the heterogenous model-based LBM simulations. The images in **Fig. 5a-c** are that in the electrolyte at 1C, 2C, and 5C, respectively. As shown in **Fig. 5a**, the concentration profiles of lithium ions in the electrolyte is relatively homogeneous. The maximum concentration difference across the whole electrode is about 500 mol m^{-3} (25% of initial lithium ion concentration in the electrolyte) and the concentration of lithium ions near the cathode collector is 1736 mol m^{-3} (87%), namely, there is a slight concentration gradient in the electrolyte. Thus, we can conclude that the transport of lithium ions is not the limiting factor for battery performance at 1C. As the discharge rate increases, the difference becomes more extensive, and it is about 3000 mol m^{-3} (150%) at 5C. Meanwhile, the concentration near the cathode collector falls to 116 mol m^{-3} (5.8%), indicating significant

electrolyte depletion, which reveals the increasing importance of lithium ions transport in determining the performance. This outcome may be resulted from two reasons. First, the diffusion path is too long, and the transport of Li^+ is difficult from the separator to the cathode particles. Thus Li^+ has completely depleted nearby the cathode collector. After depletion, the active particles become inactive and increase the cell's resistance, resulting in poor capacity and power performance. Second, the nonhomogeneous particle distribution would hinder the diffusion of lithium ions (see the dashed box area), thus resulting in a longer diffusion path.

The images in **Fig. 5d-f** show the concentration profiles of Li^+ in the solid at 1C, 2C, and 5C, respectively. As seen in **Fig. 5d**, the concentration profiles of lithium ions in the solid is relatively homogeneous except in the aggregate of particles (see the bottom left and bottom right corners in the electrode). This is mainly due to the particle aggregation bringing a long diffusion path, which blocks the diffusion of lithium ions from the particle surface to the center. Therefore, as the discharge progresses, the Li^+ concentration in the particle surface increases rapidly and approaches the saturating value, causing electrochemical shielding on the particle surface and reducing the active material utilization rate. As seen in **Fig. 5e** and **Fig. 5f**, as the discharge rate increases, the concentration gradient of lithium ions in the solid becomes steeper with the denser the iso-concentration lines. It means the increasing importance of lithium ions transport in the solid. What's more, as the discharge rate increases, there is apparent heterogeneous utilization of active materials from the separator to the cathode collector. The lithium ions preferentially embed in the particles in the vicinity of the separator. This is mainly caused by the transport of Li^+ in the electrolyte. At high C-rates, the Li^+ is difficult to transport to the particles nearby the cathode collector and the active particles become inactive, which results in a significant amount of active material remaining

unutilized. In addition, it is noted that there is an apparent concentration difference on particle surfaces in the dashed box area. The exterior surface close to the separator is more active than the internal surface close to the cathode collector. This is mainly caused by the blocking effect of the non-homogeneous particle distribution on the ion transport stated above. In conclusion, controlling the particle distribution in the direction of electrode thickness to balance the Li^+ transport and reaction may be an effective method to increase the utilization rate of particles and improve the battery performance at high C-rates.

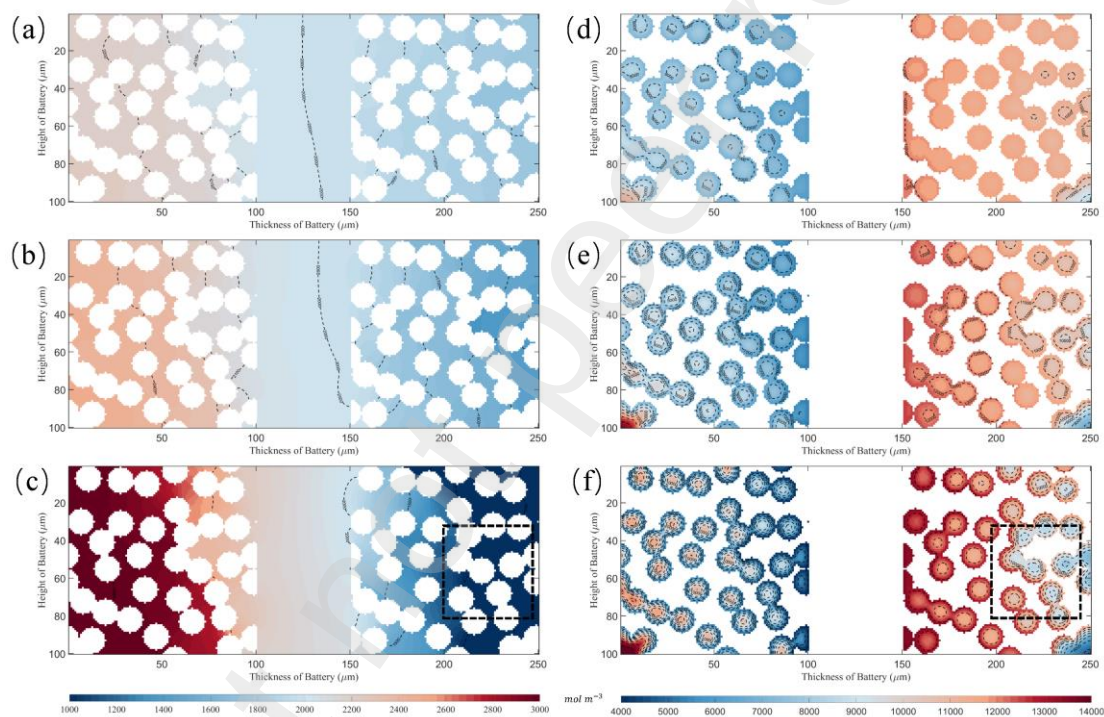


Fig. 5. Heterogeneous concentration profiles of Li^+ in the heterogeneously structured electrodes (**Fig. 3c**) at 50%

DOD in (a)-(c) the electrolyte and (d)-(f) the solid at 1C, 2C, and 5C, respectively.

3.2.2 The applicability of the tortuosity-based P2D model for heterogeneously structured electrodes

The voltage responses and the concentration profiles of Li^+ in the heterogeneously structured electrodes (**Fig. 3c**) at different discharge rates obtained by the heterogenous model are compared with that using the P2D model with differently defined tortuosity, so that the applicability of the

P2D model was investigated. The tortuosity values calculated by the Bruggemann relation and the random walk method are 1.19 and 1.52, respectively.

Fig. 6a-c compares the average lithium ions concentration distribution in the electrolyte along the direction of electrode thickness at different discharge rates, calculated using the heterogeneous model and the P2D models with tortuosity values from different methods. It can be seen that the P2D model with random walk-resulted tortuosity gives very similar concentration profiles in the electrolyte to that of the heterogeneous model at the relatively low discharge rates; while the Bruggemann relation provides fairly poor results, suggesting that the ions transport depends on electrode structure rather than only the porosity. As the discharge rate increases, there are obvious deviations in lithium ions distribution in the electrolyte (**Fig. 6b** and **Fig. 6c**), especially near the cathode collector (see the dashed box area). This outcome could be attributed to two reasons. First, the local microstructure blocks the diffusion of lithium ions, which is proved in **Fig. 5f**. Second, the ions undergo coupled diffusion-reaction processes rather than pure diffusion in porous electrodes. The electrode is usually not utilized along the whole thickness direction at high C-rates. Namely, only the tortuosity inside penetration depth works rather than the tortuosity of the entire electrode. In general, the Bruggemann relationship works poorly. It is necessary to utilize the random walk method to calculate the tortuosity because it can simulate the distribution of lithium ions in the electrolyte well. However, it doesn't work at high C-rates because it only considers the effect of physical structures on transport, while overlooking the coupling between the transport and electrochemical reactions. Moreover, as seen in **Fig. 6d**, the voltage responses curves of the P2D_Model_Brugg and the P2D_Model_RW almost coincide at low C-rates, while there is a big difference in the average lithium ions concentration distribution in the electrolyte (**Fig. 6a** and **Fig.**

6b). This mainly results by the less importance of the transport of Li^+ in the electrolyte at low C-rates. However, there is not an apparent difference at high C-rates, which indicates the macroscopic voltage responses depend on the combined action of structures, transport, and reactions. What's more, the different tortuosity-based P2D models give higher battery capacity than the heterogeneous model at all C-rates. As the discharge rate increases, the deviation becomes larger, indicating the increasing importance of local microstructure. Therefore, we may conclude that the P2D model is not able to describe exactly the physical processes inside the battery even though it may provide similar macroscopic voltage responses. In consideration of low computational cost, the P2D model is only applicable in the condition of thin and homogeneous electrode structures and at low C-rates. The heterogeneous model proposed in the paper is preferred when considering high C-rates and thick electrodes.

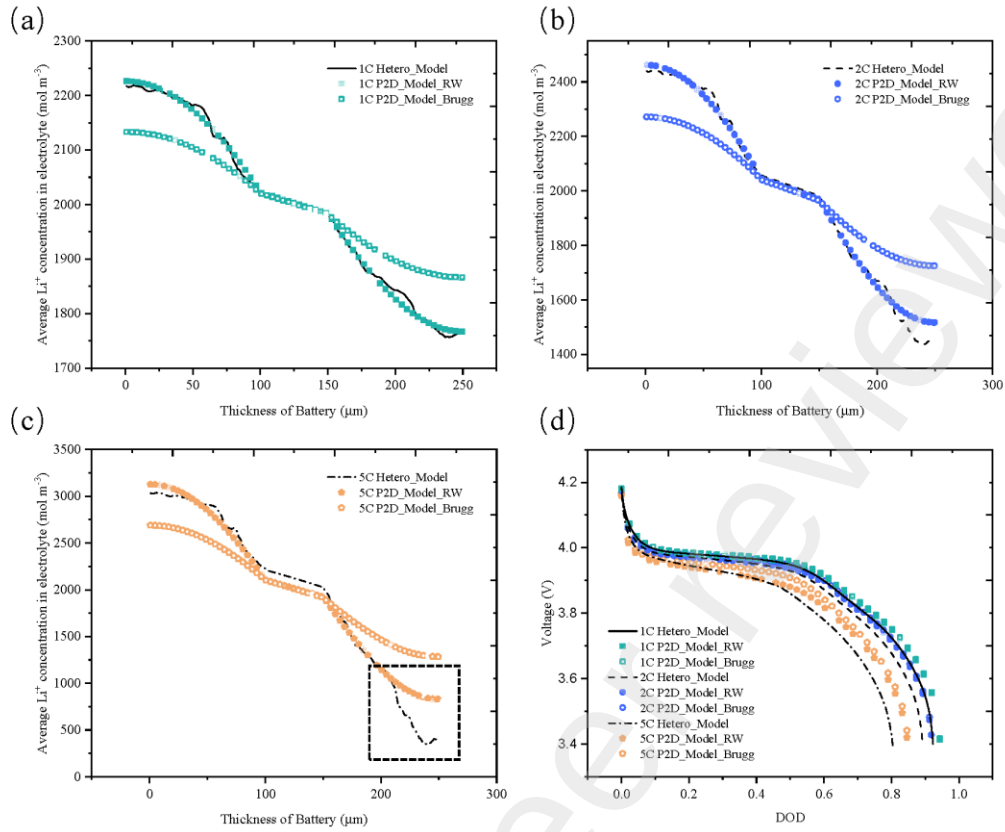


Fig. 6. The average concentration profiles of lithium ions in the electrolyte at 1C(a), 2C(b), and 5C(c) discharge using the heterogeneous model and P2D model, respectively. Results at 50% DOD are shown. The Hetero_Model (full lines) represents the results using a heterogeneous model, the P2D_Model_Brugg (hollow dots) represents the results using the P2D model with the tortuosity calculated by the Bruggemann relation, and the P2D_Model_RW (filled dots) represents the results using the P2D model with the tortuosity calculated by the random walk method. Voltage response (d).

4. Conclusion

A 2D heterogeneous LIB model based on lattice Boltzmann equations and the corresponding LBM simulation scheme are proposed to treat the influence of the microstructures of porous electrodes on battery performance under galvanostatic discharging conditions. The model is validated by the simulated voltage responses of the lithium manganese oxide/graphite battery systems composed of structure-uniform active particles homogeneously distributed in the cathode

and anode respectively. For battery systems with heterogeneously structured electrodes, the simulation results indicate that the local microstructures crucially determine the ion transport kinetics in porous electrodes and its coupling with the ion interaction reactions, therefore the utilization of the active materials, especially at high C-rates. The present results manifest the weakness of the commonly used P2D model, and emphasize the significance of microstructure design in constructing high performance LIB electrodes. It is found that the P2D model-based simulations using electrode tortuosity determined according to the real physical structures through the random walk method can well describe the transport-reaction coupling in heterogeneously structured electrodes at low and intermediate C-rates, while that using tortuosity from the Bruggemann relationship only works at low C-rates. The heterogeneous model can well describe the influence of electrode structures on battery performance at high C-rates because it fully considers the physical structure effect on the coupling between ion transport and electrochemical reactions in porous electrodes.

CRedit authorship contribution statement

Qiang Shan: Conceptualization, Methodology, Software, Investigation, Validation, Data curation, Writing- Original draft preparation. **Yuwen Liu:** Conceptualization, Methodology, Supervision, Project administration, Funding acquisition. **Shengli Chen:** Conceptualization, Methodology, Writing-review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was carried out with funding from the Natural Science Foundation of China (Grant Nos. 21773175, 21832004).

Appendix A. Supplementary data

Asymptotic expansion details for lattice Boltzmann equations, and material properties parameters in galvanostatic discharge simulations.

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